

The document provides an overview of solid rocket propellants, their types, components, and characteristics. Here is a complete summary:

Solid Rocket Propellants:

1. Requirements:

- Must have enough structural integrity to withstand pressure and g-forces.
- Should burn at a proper rate to create the necessary pressure.

2. Types of Propellants:

- **Double Base Propellants:**
 - Homogeneous mixture of fuel and oxidizer at the molecular level.
 - Common components: nitrocellulose and nitroglycerine.
 - Used in military tactical missiles.
 - Known for safety and performance (refer to safety check table 12.3).
- **Composite Propellants:**
 - Heterogeneous mixture with oxidizing crystals, a plastic-like fuel binder, and metal powders.
 - Common metals: Aluminum (Al), Magnesium (Mg), Beryllium (Be).
 - Common oxidizers: Ammonium perchlorate (AP), Ammonium nitrate (AN), Potassium perchlorate (KClO₄).
 - Binders: Hydroxyl-terminated polybutadiene (HTPB), Polybutadiene acrylonitrile (PBAN).

3. Components and Characteristics:

- **Aluminum (Al) Powder:**
 - Most common metal fuel, increases specific impulse (Isp).
- **Magnesium (Mg):**
 - Clean fuel, particle size 70-40 micrometers.
- **Beryllium (Be):**
 - Most energetic, used in aggregates.
- **Ammonium Perchlorate (AP):**
 - Moderately low cost, moderate performance, produces hydrochloric acid.
- **Ammonium Nitrate (AN):**
 - Cheaper, less energetic, clean.
- **Potassium Perchlorate (KClO₄):**
 - High-powered, used in hot gas rockets.

4. Binders:

- **HTPB:**
 - Long-chain polymer, most common today.

- **PBAN:**
 - Used in Space Shuttle Solid Rocket Boosters (SRBs), similar to rubber particles.

5. **Performance Metrics:**

- Example of Shuttle SRB:
 - Burn time: 2 minutes.
 - Pressure: 8410 kgs, 622 Mpa.
 - Specific Impulse (Isp): 292 seconds.
 - Propellant composition by mass: AP (69.6%), Al (16.0%), Fe oxide (0.4%), PBAN binder (12.04%).

6. **Health and Safety:**

- Chronic exposure to certain components like AP can cause thyroid issues.
- Safety and performance data are detailed in tables 12.2, 12.3, 12.13, and 12.17.

End-capped polymers are polymers that have specific chemical groups at the ends of their polymer chains, giving them unique properties. In the context of solid rocket propellants, two common end-capped polymers are:

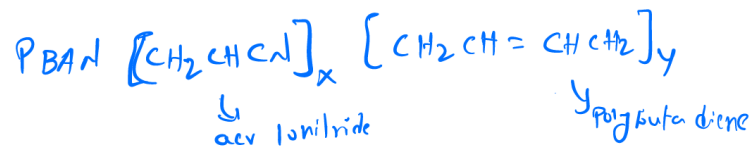
HTPB (Hydroxyl-Terminated Polybutadiene):

This polymer has hydroxyl (OH) groups at the ends of its polymer chains. These groups can react with other chemicals to form cross-linked structures, which enhance the strength and flexibility of the propellant.

PBAN (Polybutadiene Acrylonitrile):

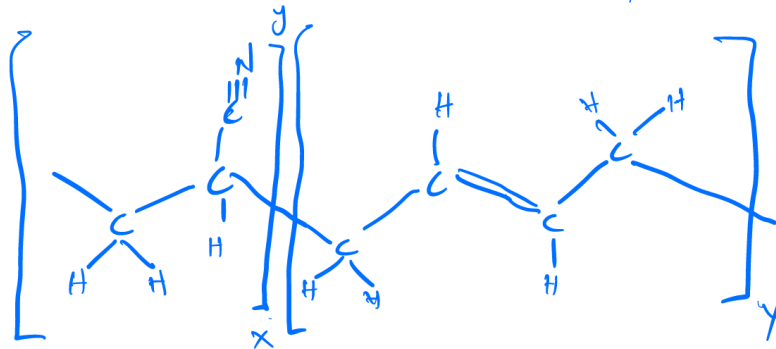
This polymer has nitrile (CN) groups at the ends of its polymer chains. These groups provide the polymer with specific properties such as heat resistance and flexibility. PBAN is used in the solid rocket boosters of the Space Shuttle.

These end-capped polymers are valuable in solid rocket propellants due to their chemical and physical properties, which help improve the performance and safety of the propellants.



mass
 F 28.7% $\rightarrow$ Al 57.1%
 OX 71.3% $\rightarrow$ PBAN 42.9%

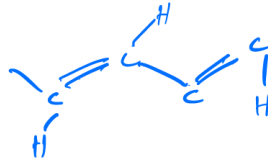
$r = \frac{\text{OX}}{\text{F}} = 2.48$



HTPB

$$\rho = 1860 \text{ kg/m}^3$$

Al/AP/HTPB



40-50 butadiene chains

terminated with -OH

in CEA at C_4H_6 , butadiene

The metal particles can get large enough

they have significantly different $v + T$ than gas phase.

Look at 12.2 shows $T_{02} + I_{sp}$

for different oxidizers + Percentages

Also look at 12.13 & 12.14